

The Utility Problem

The concept of data challenges is nothing new to electricity distribution utilities. Data challenges are special requests for the public, academia or technology providers to submit innovative solutions to real-world utility problems. This data challenge involves the processing of a wide range of datasets to obtain a solution. The ideas for solutions and the executions are usually evaluated competitively where the winner often earns some kind of reward. Utilities have been running data challenges in all shapes and forms for quite some time.

The common denominators for all utility data challenges can be summarized as follows:

1. Utility publishes a real-world problem which can be solved by processing data.
2. Utility informs challenge participants about intended datasets, often with a short description of these datasets.
3. Data challenge participants submit their ideas for solving the challenge.
4. Utility shortlists participants, so finalists get an opportunity to solve the challenge.
5. The utility collects all datasets which are intended for the data challenge.
6. The utility explains the data and data relations to finalists.
7. Data challenge finalists work on the solution to the challenge.
8. Solutions to the challenge are evaluated and winners are announced.

Some utilities conduct data challenges quite often, so the above-mentioned steps are repeated again and again. However, regardless of the fact of how often the data challenge is conducted, it is essential to minimize the manual steps in this process and make the process as streamlined as possible. Such improvements will add credibility to the utility running the challenge as well as attract more high-quality challenge participants so that the winning solutions can be the best possible.

The Data Solutions and Methods

Since the Awesense Platform and its Energy Data Model (EDM) interface are proven tools for addressing a wide range of utility use cases, the mentioned tools can also be swiftly applied for the purposes of running the utility data challenge in the form of a utility data challenge hub. To explain where the Awesense Platform and the EDM can be beneficial, let's comment on the above-mentioned steps.

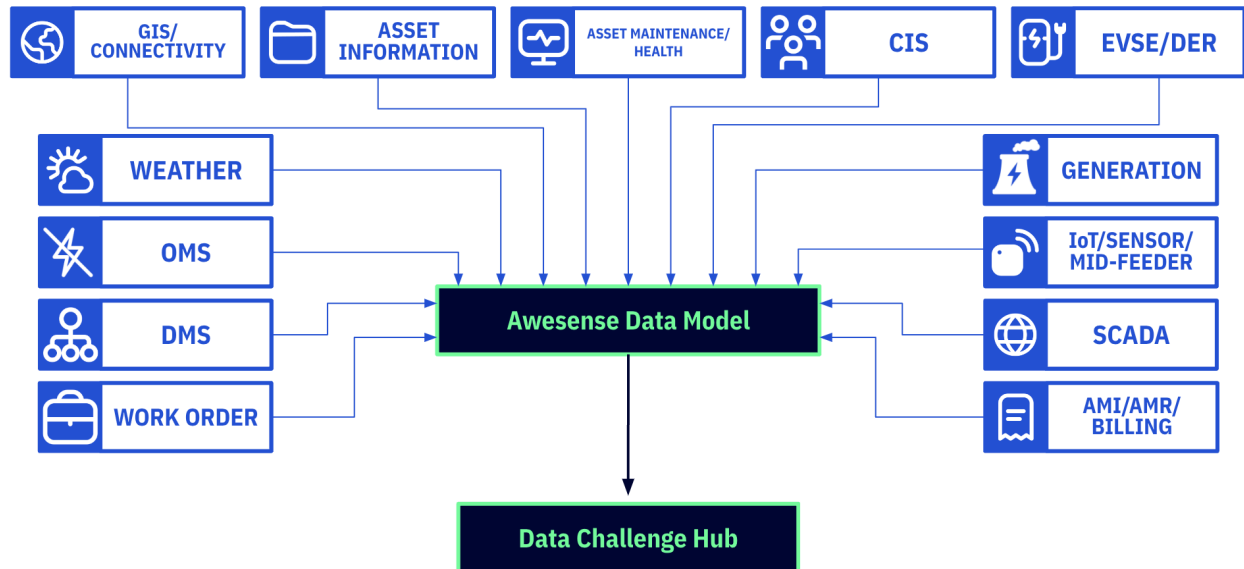
1. Utility publishes a real-world problem which can be solved by processing data.
 - a. In addition, utilities might gain inspiration from Awesense's published set of use cases which can be solved by the Awesense Platform.
2. Utility informs challenge participants about intended datasets, often with a short description of these datasets.
 - a. With the Awesense Platform, informing about available datasets is easy. This can be achieved programmatically by calling API functions.
3. Data challenge participants submit their ideas for solving the challenge.

- a. Challenge participants might test their ideas on the Awesense sandbox environment, which contains synthetic but realistic utility datasets. This might help them to propose better ideas.
4. Utility shortlists participants, so finalists get an opportunity to solve the challenge.
5. The utility needs to collect all datasets which were intended for the data challenge.
 - a. This is not necessary to do since all required data is already ingested and available within the Awesense Platform.
6. The utility will spend time with the finalists' participants to explain the data and the relations between the data.
 - a. Again, this is not necessary to do since all data semantics are well explained in Awesense Platform documentation.
7. Data challenge finalists will work on the solution to the challenge.
 - a. Building the solution for the challenge against the Awesense EDM is easier for multiple reasons:
 - i. The data is already cleaned and synchronized by Awesense Dataengine (using VEE - validation, estimation and editing),
 - ii. Awesense Platform provides out-of-the-box data visualizations for data exploration,
 - iii. Awesense EDM data views and functions simplify work with complex utility data,
 - iv. Analysts can gain inspiration from Awesense's published analysis tutorials and best practices.
8. Solutions to the challenge are evaluated and winners are announced.
 - a. Among the advantages of the solution which is built against the Awesense Platform is the fact that the winning solution can be easily adapted by the utility or easily ported to a sister utility if the sister utility has the Awesense Platform implemented as well.

From the comments above, it is clear that using the Awesense Platform as the data hub for utility data challenges is beneficial for saving the time of the utility personnel as well as for building the solution to the utility challenge easier.

Required Data Sets

The following graphics show all datasets which can be ingested to the Awesense Platform and consequently exposed for the purposes of the Utility Data Challenge.



Available Use Case Implementations:

Users of the Awesense Platform can access this use case by the following means:

1. Please contact [Awesense](#) if you would like to see a demo of this use case.